metals. It is a combination of two elements like Iron and carbon. Other elements are present in small qualities. As carbon content rises the metal becomes harder and stronger but less ductile and more difficult to weld. Carbon steels are further classified into three on the basis of percentage of carbon.

1. Low carbon steels or mild steels contain up to 0.3%of carbon.
2. Medium carbon steels contain 0.3% to 0.6% carbon.
3. High carbon steels contain 0.6% to 1.5% of carbon.

Alloy Steels: Alloy steels contain alloying elements (Example Manganese, silicon, nickel, titanium, copper, chromium and aluminum) in varying proportional in order to manipulate the steel's properties, such as it's harden ability corrosion resistance, strength, weldability or ductility.

a) **Stainless Steel:** Stainless steel generally contain 10-20% of chromium as the alloying elements. It is notable for its corrosion resistance, and it is widely used for food handling and cutlery devices.

b) **Tool Steel:** Tool steel contain tungsten, molybdenum, cobalt and vanadium in varying quantities to increase heat resistance and durability, making them ideal for cutting and drilling equipment.

Examples of Tool Steel:

High speed Steel: High speed steel is a cutting tool material used in drilling, milling, turning, threading, boring, gear cutting and many other machining operations. The alloying elements are tungsten, molybdenum, vanadium and chromium.

Tungsten Steel: High speed Tungsten steel is a high carbon tool steel, containing a large dose of tungsten; a typical HSS composition is 18% tungsten, 4% chromium, 1% vanadium, 0.7% carbon and the rest, Iron.

Cast Iron: Cast iron is a group of iron-carbon alloy with carbon content ranges from 2% to 4.5%. This high carbon content makes them excellent materials to use for casting.

Types of cast Iron

1. White Iron
2. Gray Iron
3. Ductile Iron
4. Malleability Iron

Non Ferrous: Non-ferrous metals do not contain Iron, we can also get non-ferrous metals as alloys. Eg, brass is an alloy of copper and Zinc. Non-ferrous metals are specified for structural applications because it has lesser weight, higher strength, higher melting points. They are also used for electronic applications. Some examples of Non-Ferrous metals are given below

1. Aluminium and Aluminium Alloys
2. Copper
3. Brass
4. Lead
5. Zinc

Aluminium: Aluminium is a chemical element with symbol Al and Its atomic number is 13. It is a silvery-white, soft, non-magnetic, ductile metal. Its ore is bauxite. As pure Aluminium has a low melting point(658°C)and it is used for sand casting.

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